

Bellabeat Fitness — SQL Analysis & Proof

Author: Varshith · Queries executed against the Fitbit dataset (33 users, ~31 days)

This document contains each SQL query used to clean and analyse the Bellabeat (Fitbit) data, the actual result it returns, and a short interpretation.

1. User coverage by table

```
SELECT 'daily_activity' AS source, COUNT(DISTINCT Id) AS users
FROM daily_activity
UNION ALL SELECT 'sleep_day', COUNT(DISTINCT Id)
FROM sleep_day
UNION ALL SELECT 'weight_log', COUNT(DISTINCT Id)
FROM weight_log
```

source	users
daily_activity	33
sleep_day	24
weight_log	8

Insight: Activity covers all 33 users; sleep 24; weight only 8 - weight conclusions are weak.

2. Daily averages (steps, calories, sedentary minutes)

```
SELECT ROUND(AVG(TotalSteps)) AS avg_steps, ROUND(AVG(Calories)) AS avg_calories,
ROUND(AVG(SedentaryMinutes)) AS avg_sedentary_min
FROM daily_activity
WHERE TotalSteps>0
```

avg_steps	avg_calories	avg_sedentary_min
8319.0	2361.0	956.0

Insight: Users average ~8,300 steps and ~16 sedentary hours per day.

3. % of days below the 10,000-step goal

```
SELECT ROUND(100.0*SUM(CASE WHEN TotalSteps<10000 THEN 1 ELSE 0 END)/COUNT(*),1) AS
pct_below_10k FROM daily_activity
WHERE TotalSteps>0
```

pct_below_10k
64.9

Insight: Nearly two-thirds of days fall short of 10k steps.

4. User segments by average daily steps

```
SELECT activity_level, COUNT(*) AS users
FROM (SELECT Id, CASE WHEN AVG(TotalSteps)<5000 THEN 'Sedentary' WHEN AVG(TotalSteps)<7500 THEN 'Lightly
Active' WHEN AVG(TotalSteps)<10000 THEN 'Fairly Active' ELSE 'Very Active'
END AS activity_level
FROM daily_activity
WHERE TotalSteps>0
GROUP BY Id)
```

GROUP BY activity_level
ORDER BY users DESC

activity_level	users
Fairly Active	10
Lightly Active	9
Very Active	7
Sedentary	7

Insight: Users split fairly evenly across activity levels.

5. Average steps by day of week

SELECT DayOfWeek, ROUND(AVG(TotalSteps)) AS avg_steps
FROM daily_activity
WHERE TotalSteps>0
GROUP BY DayOfWeek
ORDER BY avg_steps DESC

DayOfWeek	avg_steps
Tuesday	8949.0
Saturday	8947.0
Monday	8488.0
Thursday	8185.0
Wednesday	8158.0
Friday	7821.0
Sunday	7627.0

Insight: Activity is steady; Tuesday and Saturday are most active.

6. Sleep summary and efficiency

SELECT ROUND(AVG(TotalMinutesAsleep)) AS avg_asleep_min, ROUND(AVG(TotalTimeInBed)) AS avg_in_bed_min,
ROUND(100.0*SUM(TotalMinutesAsleep)/SUM(TotalTimeInBed),1) AS efficiency_pct FROM sleep_day

avg_asleep_min	avg_in_bed_min	efficiency_pct
419.0	458.0	91.4

Insight: ~7 hours asleep with 91% efficiency - the gap is bedtime, not restlessness.

7. % of nights under 7 hours of sleep

SELECT ROUND(100.0*SUM(CASE WHEN TotalMinutesAsleep<420 THEN 1 ELSE 0 END)/COUNT(*),1) AS pct_under_7hr
FROM sleep_day

pct_under_7hr
44.1

Insight: About 44% of nights fall short of the 7-hour guideline.

8. Peak activity hours (avg steps by hour)

SELECT Hour AS hour_of_day, ROUND(AVG(StepTotal)) AS avg_steps
FROM hourly_steps

GROUP BY Hour
ORDER BY avg_steps DESC LIMIT 5

hour_of_day	avg_steps
18	599.0
19	583.0
17	550.0
12	549.0
14	541.0

Insight: Activity peaks in the early evening (5-7pm) and at noon.

9. Device usage frequency

SELECT ROUND(AVG(days_used),1) AS avg_days_used
FROM (SELECT Id, COUNT(DISTINCT ActivityDate) AS days_used
FROM daily_activity
GROUP BY Id)

avg_days_used
28.5

Insight: Users wear the device ~28 of 31 days - very high engagement.

10. Weight-logging adoption

SELECT COUNT(DISTINCT Id) AS users_logging, SUM(CASE WHEN IsManualReport='True' THEN 1
ELSE 0 END) AS manual_logs, COUNT(*) AS total_logs
FROM weight_log

users_logging	manual_logs	total_logs
8	0	67

Insight: Only 8 of 33 users log weight, mostly manually - a clear engagement gap.

Conclusion & Marketing Recommendations

Users are highly sedentary (~16 hrs/day) and miss the 10,000-step goal on roughly two-thirds of days. Activity peaks at noon and in the early evening (5-7pm). Sleep averages ~7 hours but ~44% of nights fall short, with the gap being late bedtimes rather than poor sleep quality. Engagement is high (device worn ~28 of 31 days), yet only 8 of 33 users log weight. Recommended Bellabeat marketing actions: movement reminders and personalised step goals; notifications timed to the noon and evening activity peaks; bedtime / sleep-coaching features; effortless weight logging; and streak and reward features that build on the already-high daily engagement.